Lazizbek, MATH-464. HW #5.

Prove the following:

Consider the standard form polyhedron

$$P = \{ x \in \mathbb{R}^n \mid Ax = b, \ x \geq 0 \}$$ and

assume that rows of matrix A are linearly independent. If two different bases correspond to the same basic solution, then this basic solution is <u>degenerate</u>.

Solution:

Before I start the proof, let me introduce a useful definition that I'll be using later.

Definition 2.11.

Let x be a basic solution of a standard form polyhedron with m equality constraints. Then x is a <u>degenarete basic solution</u> if more than $(n-m)$ of the components of x are zero.

Proof.

Let the standard polyhedron P,

(1) $P = \{ x \in \mathbb{R}^n \mid Ax = b, \ x \geq 0 \}$ and

(2) $\text{rank}(A) = m$ i.e. rows of A are lin. indep. (L.I.)

Let

(3) $\beta = \{B(1), B(2), \ldots B(m)\}$ &

(4) $\gamma = \{B(1), B(2), \ldots, B(m-1), B(m+1)\}$

Also, let

(5) $B = [A_{B(1)}, A_{B(2)}, \ldots A_{B(m-1)}, A_{B(m)}]$ be

a basis matrix corresponding to basis β, &

(6) $C = [A_{B(1)}, A_{B(2)}, \ldots, A_{B(m-1)}, A_{B(m+1)}]$ be

a basis matrix corresponding to basis γ.

Note that we have been given

(7) $Bx = b$ has a solution x^* &

(8) $Cx = b$ has the same solution x^*.

That is

$$[A_{B(1)} \; A_{B(2)} \ldots A_{B(m-1)}, A_{B(m)}]\, x = b \quad \Rightarrow$$

(9) $x^* = [x_1, x_2 \ldots x_{m-1}, x_m, 0, 0, \ldots 0]^T$

Based on the procedure to find a basic solution, basic variables $(x_1, x_2 .. x_m)$ are not zero, all other non basic variables $(x_{m+1}, .. x_n)$ are zero, because $\text{rank}(A) = m$.
Similarly for $Cx = b$

$$\left[\dot{A}_{B(1)}, \dot{A}_{B(2)} \dots \dot{A}_{B(m-1)} \quad A_{B(m+1)} \right] x = b \Rightarrow$$

$$(10) \quad x^* = [x_1, x_2, \dots x_{m-1}, 0, x_{m+1}, 0, \dots 0]^T$$

As x^* in (10) is the same with x^* in (9), we can equalize them as follows:

$$x^* = x^* \iff$$

$$(9) \rightarrow [x_1, x_2, \dots, x_{m-1}, x_m, 0, 0, 0 \dots 0]^T =$$

$$(10) \rightarrow = [x_1, x_2, \dots, x_{m-1}, 0, x_{m+1}, 0, 0, \dots 0]^T$$

Componentwise equality of (9) & (10) gives

$$(11) \begin{cases} x_1 = x_1, \; x_2 = x_2, \dots x_{m-1} = x_{m-1}, \; x_m = 0, \; x_{m+1} = 0, \\ x_{m+2} = 0, \dots x_n = 0 \end{cases}$$

Hence,

$$(12) \quad x^* = [\underbrace{x_1, x_2, \dots x_{m-1}}_{(m-1) \text{ basic variables } \neq 0}, \underbrace{0, 0, \dots 0}_{(n-(m-1)) \text{ nonbasic variables } = 0}]^T$$

So, our basic x^* solution for the given standard form polyhedron has $n-(m-1) = n-m+1$ zero components.

Since $n-m+1 > n-m$, by Definition 2.11, I gave on page 1, x^* is a degenerate basic solution.

$\blacksquare$.